# Kexin Shang

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## EDUCATION

**Drexel University, PA** Sep 2023 - Present
Doctor of Philosophy in Information System **GPA: TBD**
(Focus on NLP in Healthcare)

**Washington University in St. Louis, MO** Dec 2021
Master of Science in Biostatistics and Data Science **GPA: 3.94**

**Georgia State University, GA** Jun 2019 - May 2023
Bachelor of Science in Mathematics (Statistics) **GPA: 3.85**
Bachelor of Science in Biology (Second Major)

**Southwest Jiaotong University, China** Sep 2017 - Jun 2019
Bachelor of Engineering in Bioengineering **GPA: 3.73**

Main courses: *Mathematical Statistics, Biostatistics (SAS), Principles of Computer Science (JAVA), Python Programming, Analysis, Optimization, Methods of Regression and Analysis of Variance, Survival Analysis, Data Mining (R), Bioinformatics, Applied Deep Learning Using Health Data (Python), Randomized Controlled Trials, etc.*

## INTERNSHIP EXPERIENCE

**Center for Healthy Weight and Wellness, Psychiatry, WUSTL** Intern May – Dec 2022
**(Collaborated with PSU, UCLA, Umich, and Stanford)**
Topic: Harnessing Mobile Technology to Reduce Mental Health Disorders in College Population

- Constructed composite variables from over 200 features via PCA regression, which determines nearly 40% of the variation of response rate to follow-up surveys
- Based on a factorial design on 4 treatment components, identified moderator variables within each component using logistic regression models and simple slope analyses
- Conducted pairwise T-tests and Chi-square tests with Holm's corrections to compare 11 types of clinical and subclinical eating disorders in rural areas, suburban areas, and urban areas of the U.S.

**Zhang Lab, Department of Developmental Biology, WUSTL** Research Assistant Sep 2021- May 2022
Topic: The Role of Transposon in Transcript-level Expression Regulation

- Developed a Shell-based pipeline to obtain TE-derived transcripts' expression contribution from GTEx database
- Located age-sensitive TE-derived transcripts in skin tissue by plotting time-series Z-scores across age intervals
- Removed unwanted variation using residuals (RUVr with k=4) from RNAseq data and plot a 3D PCA (after RUVr), which successfully showed clear separations between sun-exposed skin genes and sun-unexposed skin genes

**Department of Orthopaedical Surgery, WUSTL** Research Assistant Jun - Jul 2022
Topic: Senior Athlete Pain Study

- Performed inferential statistical analysis based on a cross-sectional survey of senior athletes in the U.S.
- Plan to characterize biopsychosocial phenotypes of senior athletes via unsupervised latent class analysis

## IN-SEMESTER PROJECTS

**MSB 660-01 Biomedical Data Mining** WUSTL 2022 Spring
Topic: Medical Cost Prediction

- Leveraged the Medical Expenditure Panel Survey (MEPS) database to predict medical cost across 3376 patients by fitting models of multiple linear regression, bagged random forest regression, and logistic regression w/t lasso penalty
- Adopted LDA and Naïve Bayes classifier to classify patients with high medical cost, achieving 96.9% and 94.1% specificity respectively
- Used Inverse normal transformation (INT) to normalize highly skewed data (change skewness from 4.9 to 0.074)

**BMI 5303 01 Introduction to Biomedical Informatics II** WUSTL 2022 Spring
Topic: Correlation of No-Mammogram Rate vs Breast Disease Prevalence at County Level in Missouri

- Collected and cleaned data of county-level population and mammogram rate from Missouri Census Data Center and Missouri Department of Health & Senior Service

- Used MDClone, a synthetic data platform, to generate simulated patient cohort of breast disease in Missouri
- Merged MDClone data to county-level census profile, fitting a robust exponential regression (R2 = 0.7) curve of no-mammogram rate and breast disease prevalence

**MATH 4544 Biostatistics** GSU 2019 Spring

Topic: How Infants' Blood Pressure Effluences Their Ability to Suck Saline

- Applied *Descriptive Statistic Kit* in EXCEL to calculate descriptive statistics of diastolic pressure (SDP) and systolic pressure (DBP)
- Performed ANOVA test to compare the mean and 95% CI of sucking ability of different SDP and DBP interval

## CONFERENCE

2022 ASA Women in Statistics and Data Science Conference, St. Louis, MO — Audience

2023 International Conference on Eating Disorders (ICED), Washington, DC — Poster Presenter

## HONOURS & AWARDS

Valedictorian of the Recognition Ceremony, WUSTL — 2022

Merit Scholarship ($11,886), WUSTL — 2021

Wiley M. Suttles Math Award ($750), GSU — 2023

In-state Scholarship; Presidential List; Member of the Honors College, GSU — 2020-2021

Second-class Comprehensive Scholarship (¥3000); Merit Student; National Volunteer, SWJTU — 2018-2019

## EXTRACURRICULAR ACTIVITY

Publicity Department of the Chinese Student Union, GSU — Minister

- 2020 Atlanta Chinese Students and Scholars Spring Festival Gala
- Social media account management and operation

## SKILLS

Analytics: Machine Learning Models, Artificial Intelligence with Popular APIs (fastai2, Hugging Face, Haystack, Google Colab, Kaggle), Deep Learning Architectures (CNN, RNN, transformers) on various data types (text, image, video, audio, and tabular)

Programming: R, SAS, Python (Pandas, Numpy, Tensorflow, Pytorch), Shell, MySQL, Latex, MDClone

Language: English (fluent); Chinese (native)

Certification: Machine Learning by Stanford University on Coursera